
Closing the Causal Loop: Public and Private Routes in Autoregressive Models

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Mechanistic interventions on autoregressive language models are often interpreted
2 as evidence that internal state directly controls later tokens. But generation closes
3 a causal loop: an intermediate output becomes the model’s next input, so the
4 same downstream change can arise either from contextual influence that persists
5 after the intermediate token is fixed or from feedback through the model’s own
6 generated token. We introduce a six-cell intervention assay and a four-stage causal-
7 validity ladder that separate these public and private routes. On an occupation-
8 completion task adapted from prior latent-planning work, an attribution-selected
9 sparse intervention in Gemma 3 270M with both article and future-noun influence
10 rewrites completions such as *a pilot* to *an aviator* through an almost entirely public
11 route: replaying the treated article reproduces the noun, restoring the baseline article
12 rescues it, and article substitution reproduces almost all of the noun-distribution
13 change. Yet target-directed residual interventions still shift later nouns with the
14 article fixed at both 270M and 1B, demonstrating private-route capacity. A neutral-
15 prompt synonym assay at 1B then reveals mediator-relative routing: when two valid
16 alternatives require different articles, target-directed influence is public-dominant;
17 when they share the same article, it is private-dominant. In a matched 14-family
18 extension holding semantic family and source state fixed, crossing the article
19 boundary shifts route dominance toward the public path in 13/14 families. Finally,
20 for the binary article mediator, public effect magnitude factorizes exactly into
21 mediator-policy movement and continuation leverage: near an article decision
22 boundary, crossing the boundary produces a large noun-distribution jump while
23 either fixed-article path moves very little. Future-token control is therefore a closed-
24 loop causal quantity: mechanistic claims must identify not only what internal state
25 changes, but where that influence travels and how generation amplifies it.

26 1 Introduction

27 Language-model outputs are usually treated as readouts of internal computation. In an autoregressive
28 model they are also interventions on the model’s future: each sampled token is appended to context
29 and processed again. This feedback creates two causal routes from an internal state to a later
30 prediction. Influence can remain **private**, propagated through contextual hidden state, or become
31 **public**, projected into a generated symbol and returned as a high-leverage input. A later-token change
32 observed under free generation does not say which route carried it.

33 This ambiguity is acute in work on latent planning. Hanna and Ameisen define it as an internal
34 representation that both causes a specific future token or concept and shapes preceding context to
35 license it [Hanna and Ameisen, 2026]. Those criteria do not distinguish whether the future effect
36 persists privately or is transmitted through the preceding context the representation caused. Thus a
37 rewrite from a *pilot* to an *aviator* could reflect persistent noun control or a closed loop in which

38 changing a to an changes the next prediction. Our question is not whether such a signal is causal, but
 39 how its influence travels.

40 We develop an intervention framework that separates them. A six-cell assay crosses intervention
 41 off/on with free generation, forced a, and forced an. It decomposes the downstream probability-vector
 42 change into a component reproduced by public-token substitution and a matched-token component
 43 that persists privately. A four-stage **causal-validity ladder** additionally requires that a candidate
 44 signal be locally causal, that the mediator states induced by the intervention lie within the assay’s
 45 support, and only then that its transmission route be interpreted.

46 The experiments reveal a coherent organization of causal state.

- 47 1. **Public projection can dominate apparent future-token control.** In Gemma 3 270M, an
 48 attribution-selected sparse intervention produces a 0.893-TV noun rewrite, of which 0.891
 49 is reproduced by article substitution; only 0.028 TV remains with the treated article fixed.
 50 Article replay and rescue recover the corresponding top noun on every one of 20 held-out
 51 prompts.
- 52 2. **Private target information remains causally effective.** Natural-state-derived residual
 53 patches move target nouns while article identity is fixed at both 270M and 1B. Public
 54 mediation is therefore not the only available route.
- 55 3. **Routing is mediator-relative.** In a neutral-prompt 1B synonym assay, both lexical alterna-
 56 tives remain correct. In a matched 14-family assay holding semantic family and source state
 57 fixed, crossing the article-equivalence boundary shifts causal influence toward the public
 58 route in 13/14 families; an independent-family assay shows the corresponding public/private
 59 double dissociation in all 14 families.
- 60 4. **Public effects have closed-loop gain.** Once a distinction enters the public route, its
 61 downstream magnitude is the product of mediator-policy movement and continuation-
 62 branch leverage. At 270M, composing local causal pressure with decoder susceptibility
 63 and leverage provides limited predictive validation in the decisive-policy regime, whereas
 64 attribution alone does not; prediction among small effects remains weak.

65 The resulting claim is not that hidden computation is absent, nor that the model spontaneously
 66 chooses between two independent memories. Rather, a generated token is a **lossy public projection**
 67 of future-relevant state. Distinctions visible in that projection can acquire a public causal route;
 68 distinctions collapsed by it cannot travel through token identity and remain target-aligned through
 69 private context in our controlled assay. Mechanistic claims about future tokens must therefore identify
 70 both **routing**—where the causal influence travels—and **gain**—how the decoding loop amplifies it.

71 This extends prior distinctions between decodable and causally used planning signals [Ma and Rui,
 72 2026], and complements work showing that hidden state beneath fixed visible scratchpad tokens can
 73 matter [Shih et al., 2026]. Generated-token mediation itself is not new [Vig et al., 2020, Paul et al.,
 74 2024]. Our contribution is to turn it into an operational validity standard for internal mechanistic
 75 interventions and to show public dominance, private persistence, and mediator-relative route reversal
 76 within one controlled setting. Transmission route is orthogonal to functional interpretation: a state
 77 may satisfy a latent-planning definition while its future effect is public, private, or hybrid.

78 2 Public projection and private persistence

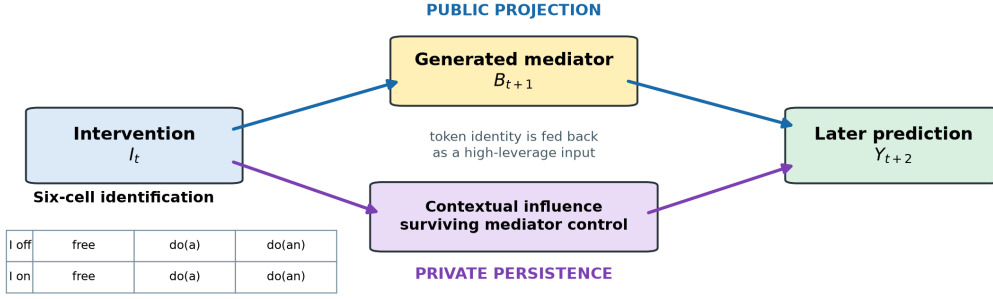
79 Let X end immediately before a generated mediator, let $B \in \{a, an\}$ be its public token identity, and
 80 let Y be the next-token distribution over the first noun token. An internal intervention I acts at the last
 81 position of X and remains active during noun prediction. Its effect can reach Y through two paths
 82 (Figure 1 A): by changing B , which is appended and re-embedded, or by changing contextual state
 83 available after B is fixed. We call these public projection and private persistence. The names refer to
 84 experimentally separable routes, not independent registers or claims about semantic encoding.

85 We use *projection* in the functional sense of a discrete public readout, not to imply a linear projection
 86 operator. The private branch denotes contextual influence from the earlier intervention that survives
 87 mediator control; it does not posit or localize a single hidden register beneath the generated article.

88 2.1 Six cells identify transmission

89 For every prompt we evaluate

A Autoregressive generation closes a causal loop



B Four distinct claims require four distinct checks

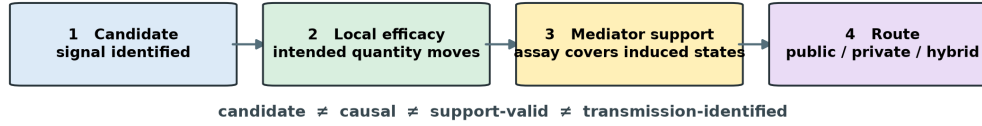


Figure 1: Autoregressive feedback and the causal-validity ladder. A: an upstream intervention can affect a later prediction through public mediator identity or private contextual state. Fixing or replaying the mediator separates the routes. B: candidate identification, local efficacy, mediator-support validity, and route identification are distinct claims.

$$I \in \{0, 1\} \times B \in \{\text{free}, \text{do}(a), \text{do}(an)\}.$$

Free generation is a policy rather than a third assigned value of B . Let B_0, B_1 denote the greedy articles under intervention off/on and let $Y(i, b)$ be the noun distribution. Define

$$\begin{aligned} E_{\text{total}} &= Y(1, B_1) - Y(0, B_0), \\ E_{\text{public}} &= Y(0, B_1) - Y(0, B_0), \\ E_{\text{private}} &= Y(1, B_1) - Y(0, B_1). \end{aligned}$$

These are full probability vectors and satisfy $E_{\text{total}} = E_{\text{public}} + E_{\text{private}}$ exactly. Their total-variation norms are not additive, so we report each norm, vector alignment, and numerical reconstruction rather than a scalar “percent mediated.” E_{public} is a treated-token replay contrast. E_{private} is what remains under the fixed treated article; it is not Pearl’s natural direct effect. We also measure the matched contrast under the other article.

Under a stochastic mediator policy with support $\mathcal{B}$,

$$E_{\text{public}} = \sum_{b \in \mathcal{B}} [q_1(b) - q_0(b)] Y(0, b), \quad q_i(b) = P(B = b \mid I = i).$$

The policy-weighted private term and exact total identity are given in Appendix A.2.

For the binary article policy this becomes

$$E_{\text{public}} = (q_1 - q_0)(Y_{an} - Y_a), \quad TV(E_{\text{public}}) = |q_1 - q_0| TV(Y_{an}, Y_a). \quad (1)$$

Equation 1 is an algebraic identity. Its empirical use is to separate local policy movement from the leverage of the continuation branches.

2.2 A causal-validity ladder

A downstream change supports a route claim only after four checks (Figure 1 B):

- 104 1. **Candidate signal identified.** A probe, attribution graph, sparse feature, or residual contrast
105 nominates a signal; these are not equivalent claims that a semantic representation exists.
- 106 2. **Locally causal.** The intervention moves the quantity it was selected to affect on held-out
107 examples.
- 108 3. **Mediator-support valid.** The assay’s mediator state space covers states induced by the
109 intervention. If called outranks both articles, a binary a/an decomposition is incomplete
110 rather than the intervention intrinsically invalid.
- 111 4. **Transmission identified.** Only a complete free/fixed assay distinguishes public, private,
112 hybrid, and negligible effects.

113 This ladder extends the established separation of decodability from causal use [Ma and Rui, 2026] by
114 making intervention validity and downstream route identification explicit. It also aligns activation
115 patching, sparse steering, and causal scrubbing with a common causal-abstraction view [Geiger et al.,
116 2025, Chan et al., 2022], complementing calls to match mechanistic evaluations to explicitly specified
117 causal mediators [Mueller et al., 2026].

118 2.3 Mediator-relative distinguishability

119 Let $m(y)$ be the public mediator licensed by lexical outcome y , and define $y_1 \sim_m y_2$ when $m(y_1) =$
120 $m(y_2)$. Teacher and educator lie in different article classes (a versus an), whereas doctor and
121 physician lie in the same class (a). A cross-class distinction is visible in article identity; a within-
122 class distinction is collapsed by it. This motivates a causal prediction: when two valid lexical
123 alternatives cross mediator classes, an intervention can recruit a public relay; when they remain in one
124 class, target-specific influence cannot be conveyed by that token identity and must remain elsewhere
125 if it survives. This is a mediator-relative routing hypothesis, not a general bit-capacity law.

126 3 Experimental design

127 3.1 Models, prompts, and inference

128 We study pretrained Gemma 3 270M and 1B [Gemma Team, 2025]. Sparse experiments use the
129 matching Gemma Scope 2 pretrained affine cross-layer transcoders [Google DeepMind, 2025],
130 building on transcoder feature-circuit methods [Dunefsky et al., 2024, Ameisen et al., 2025], through
131 circuit-tracer [Hanna et al., 2025] and NNsight [Fiotto-Kaufman et al., 2025]. The primary
132 prompts and occupation definitions follow the released materials of *Latent Planning Emerges with*
133 *Scale* [Hanna and Ameisen, 2026]; synonym families are screened from the same occupation domain.
134 Appendix A.1 records model, dictionary, and package snapshots and maps each claim to its frozen
135 configuration and machine-readable result artifact.

136 Unless stated otherwise, we recompute the full prompt with caching disabled, use bfloat16 model
137 inference, and measure full-vocabulary next-token distributions. Sparse interventions multiply
138 prompt-specific feature activations at the pre-article position. Residual interventions add a source-to-
139 target direction at a specified decoder layer. Development prompts select features, layers, or strengths;
140 headline estimates use disjoint held-out prompts or leave-one-semantic-family-out selection. Intervals
141 are 10,000-sample prompt- or family-bootstrap 95% intervals. Exact paired sign and sign-flip tests,
142 or independent-group label permutations, are used where family counts permit.

143 3.2 Public endpoint: the sparse S1 intervention

144 At 270M, attribution graphs on eight selection occupations nominate four S1 features with positive
145 article and future-noun paths. The feature set is frozen before evaluation on 20 held-out occupations
146 and amplified $5\times$. We run the six cells, then two direct falsification tests: **phenocopy**, which replays
147 the treated article with S1 off, and **rescue**, which restores the baseline article while S1 remains
148 on. A prompt-aligned binary search brackets each article argmax transition and compares free and
149 fixed-article noun distributions over the same gain interval. Float32 final-head and stochastic-policy
150 audits test whether the conclusion depends on bfloat16 logits or greedy decoding.

151 3.3 Private-route references

152 To establish that later nouns remain sensitive to upstream state after article identity is fixed, we patch
153 natural-state-derived residual directions at the pre-article position. Source and target states come
154 from occupation descriptions that do not name the tested noun. A pooled development sweep selects
155 a layer; held-out target patches are compared with wrong-target, sign-reversed, and matched-norm
156 random controls. We run the same normalized construction at 270M and 1B. These are target-directed

157 full-residual capacity interventions, not a claim that the sparse S1 features naturally carry the same
 158 state.

159 3.4 Neutral synonym routing assay

160 The mediator-relative test uses six cross-article and eight within-article occupation-synonym families.
 161 For example, *teacher/educator* cross article classes, whereas *doctor/physician* do not. Both
 162 alternatives correctly answer one occupational definition. The evaluated prompt contains an unrelated
 163 occupation-format demonstration followed by the neutral definition; it contains neither tested synonym
 164 nor a first-letter instruction. Explicit donor prompts name each source or target word and define

$$\delta = h_{\text{target}}^{(\ell)} - h_{\text{source}}^{(\ell)}, \quad h_{\text{eval}}^{(\ell)} \leftarrow h_{\text{eval}}^{(\ell)} + s\delta.$$

165 Leave-one-family-out selection chooses layer 17/25 in every fold. The primary setting is na-
 166 tive strength $s = 1$ and article-policy temperature $\tau = 1$, with $s \in \{0.5, 1, 1.5\}$ and $\tau \in$
 167 $\{0.1, 0.25, 0.5, 1\}$ as robustness checks. Before route inference we require positive held-out target-
 168 minus-source movement and high retained a/an policy mass. Controls use a wrong target and the
 169 reversed direction.

170 For each family we project public and private probability vectors onto the intended target-minus-
 171 source contrast. Let P and H denote the signed public and fixed-token private effects, respectively,
 172 and define route contrast $R = P - H$. Cross-class families predict $R > 0$; within-class families
 173 predict $R < 0$. The interaction compares the two groups, while exact within-family sign flips test
 174 each simple effect separately.

175 Before any model or route measurement, we semantically curated and froze 22 candidate triads
 176 whose three words were intended as valid alternatives for one definition. The repaired matched-triad
 177 test holds semantic family, definition, source word, source donor, and evaluated prompt fixed while
 178 comparing one same-article and one cross-article target. Its route-blind admissibility screen uses only
 179 single-token lexical realization, positive fixed-target-article local efficacy, and retained article support.
 180 The prespecified paired estimand is $R_{\text{cross}} - R_{\text{within}}$, tested by a paired semantic-family bootstrap,
 181 an exact one-sided binomial sign test, and an exact paired sign-flip randomization test. Fourteen of
 182 22 frozen candidate triads pass the screen, exceeding the minimum gate of eight.

183 3.5 Closed-loop gain prediction

184 For 32 sparse features selected on disjoint occupations, attribution is calibrated to intervention-induced
 185 article-margin movement. Equation 1 then combines predicted movement with intervention-off
 186 decoder susceptibility and fixed-branch continuation leverage. We compare constant, attribution-only,
 187 susceptibility-only, leverage-only, activation-magnitude, and constant-shift baselines. Combined
 188 feature-and-prompt holdout excludes both identities from the calibration fold while permitting
 189 intervention-off quantities for the held-out prompt. A measured-margin version is reported only as an
 190 accounting oracle.

191 4 Two causal endpoints: public relay and private persistence

192 4.1 S1’s future-noun effect is predominantly public

193 At $5\times$, S1 changes the free article from a to an on 19/20 held-out prompts and changes the top
 194 noun on the same 19. Across all 20 prompts, total noun-distribution change is 0.893 TV. Replaying
 195 the treated article with the intervention off produces 0.891 TV in the same direction; the matched
 196 treated-article leftover is 0.028 [0.021, 0.035]. The exact vector reconstruction error is below 10^{-7} .
 197 Thus the large noun rewrite is almost entirely reproduced by substituting the article token.

198 The top-1 falsification is even more direct (Figure 2 A). Treated-article replay recreates the S1-on top
 199 noun on 20/20 prompts. Restoring the baseline article while leaving S1 active restores the baseline
 200 top noun on 20/20 prompts. Phenocopy and rescue residual TVs are only 0.028 and 0.030. The
 201 pilot example therefore reads as a causal sequence: S1 changes a to an; an licenses the aviator
 202 continuation; returning a restores pilot.

203 Across prompts, the vector decomposition confirms that this is not merely a top-token coincidence:
 204 public substitution reproduces nearly all of the total noun-distribution movement, whereas the matched
 205 fixed-token residual is small (Figure 2 B).

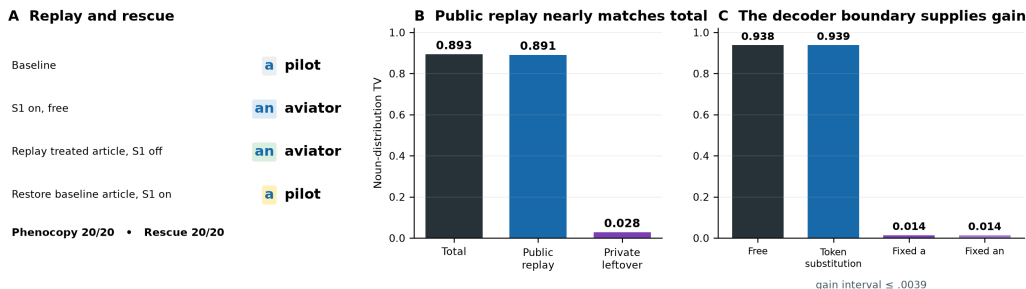


Figure 2: The sparse S1 intervention is strongly public-dominant. A: treated-article replay phenocopies the noun and baseline-article restoration rescues it. B: public token substitution reproduces almost all free-path change; TV magnitudes are shown separately and are not additive. C: around each prompt’s article boundary, the free noun distribution jumps while fixed-article paths barely move.

206 The magnitude is created at the latent-to-symbol boundary. On 19 bracketed prompts, gains separated
 207 by at most 0.0039 straddle the article argmax. The free-path noun distribution moves 0.938 TV and
 208 article substitution alone moves 0.939, with vector cosine 0.9998. Over that same interval, fixed a and
 209 fixed an distributions each move only 0.014 TV (Figure 2 C). A 16-fold tighter float32-head search
 210 yields the same qualitative result, showing that coarse bfloat16 logit levels are not the mechanism.
 211 Exact two-article mixtures from $\tau = 0.1$ to 1 replace the greedy discontinuity with graded public
 212 influence, but the policy component remains aligned with the total.

213 Thus S1 reproduces the causal pattern motivating latent-planning analyses: a pre-article handle affects
 214 both the licensing article and a future noun. A frozen-artifact audit confirms that all four features are
 215 active pre-article and have positive direct attribution to the intended future noun on all eight selection
 216 prompts. Because the features recur across nouns and selection also required article influence, S1 is
 217 not independently identified as a noun-specific semantic planning representation. Our result concerns
 218 transmission: its large free-generation noun effect travels primarily through the article, not through
 219 comparably large fixed-token persistence.

220 4.2 Target information can remain privately effective

221 The S1 result could still suggest that this task is fundamentally public: perhaps once the article is
 222 fixed, upstream state has little remaining control over the noun. We therefore ask the complementary
 223 question directly. Holding article identity fixed, can target-specific state at the pre-article position
 224 still alter the later lexical prediction? It can. A dense residual sensitivity reference first establishes
 225 reachability: on held-out same-article pairs, a target residual patch changes the target-minus-source
 226 noun logit by 6.44 [4.19, 10.06], versus 0.14 for matched-norm random directions.

227 More conservative natural-state-derived directions provide the primary private evidence. At 270M
 228 and native strength, fixed-article target-minus-source movement is 0.77 [0.47, 1.13] logits for 12
 229 cross-class pairs and 0.96 [0.67, 1.27] for 12 within-class pairs. Wrong-target effects are 0.15 and
 230 0.13; sign reversal moves the contrast in the opposite direction. The greedy article changes on
 231 0/24 pairs, so the target-directed noun movement cannot be attributed to greedy article switching.
 232 Under $\tau = 0.1$, public TV is 0.002 or less while private TV is 0.043–0.064, depending on condition.
 233 At $\tau = 1$, a graded public component also appears, so these patches are private-dominant at low
 234 temperature rather than exclusively private under every policy.

235 The target direction shifts the later noun substantially more than the wrong-target or sign-reversed
 236 controls at both scales (Figure 3 A), and the movement arises primarily by enhancing the intended
 237 target rather than merely suppressing the source (Figure 3 B). The normalized assay replicates private
 238 target influence at Gemma 3 1B. Effect sizes do not increase monotonically with scale, and we make
 239 no scale-law claim. Together with target, wrong-target, reverse, and random controls, the result
 240 establishes a second causal endpoint: target-relevant upstream state can affect the noun while visible
 241 article identity is held constant. S1 and the residual patches need not implement the same mechanism;
 242 they demonstrate that both routes are available in the model-task system.

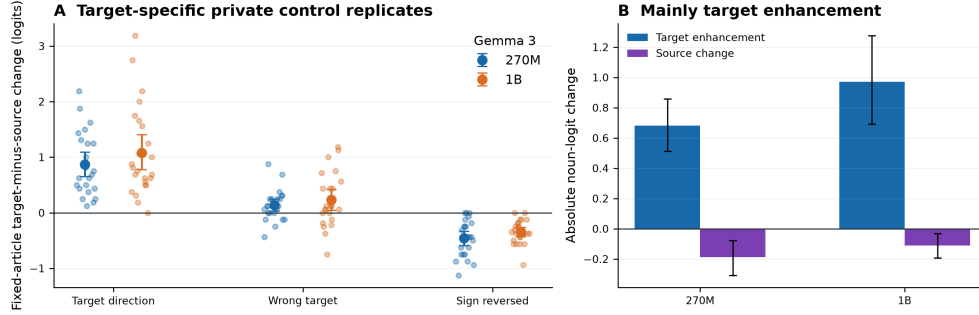


Figure 3: Target-specific private causal control. A: with article identity fixed, natural-state-derived target directions shift the target-minus-source noun contrast more than wrong-target and sign-reversed controls at both scales. B: the effect is primarily target enhancement rather than source suppression. Points are semantic families; intervals bootstrap families. These targeted patches establish private-route capacity, not a spontaneous carrier switch.

5 Mediator-relative routing

The two endpoints leave a deeper question: what determines whether an intervention recruits the public route? The neutral synonym assay compares two regimes defined by whether article identity distinguishes the intended lexical alternatives, while keeping both alternatives semantically correct and using the same intervention procedure. Concrete examples of the two mediator-relative regimes are shown in Figure 4 A.

All 14 donor-derived directions pass held-out local efficacy. At native strength, fixed-target-article target-minus-source movement is 0.406 [0.281, 0.604] logits across six cross-class families and 0.414 [0.297, 0.555] across eight within-class families. Wrong-target controls yield 0.125 and 0.008; reversed directions yield -0.385 and -0.344. Intervention-off a/an policy mass averages 0.909 (minimum 0.854) and remains 0.899 after intervention. The comparison therefore clears local efficacy and mediator-support validity without the source-initial conflict of the earlier prompt construction.

Routing then reverses by mediator equivalence class (Figure 4). For cross-class alternatives, every family increases the target-article policy; mean target-article movement is 0.182 and 2/6 switch the greedy article. At $\tau = 1$, public TV is 0.176 [0.108, 0.243] and private TV is only 0.021 [0.012, 0.033]. Their target-aligned effects are 0.189 [0.065, 0.326] and 0.019. For within-class alternatives, no family switches the article and mean policy movement is 0.009. Public TV is 0.009 [0.003, 0.018], whereas private TV is 0.061 [0.046, 0.075]. More importantly, public target-aligned influence is approximately zero -0.001 [-0.003, 0.001], while private target-aligned influence is 0.093 [0.062, 0.124].

A matched-triad test asks whether the reversal could reflect between-family semantic composition (Figure 4 B). Fourteen neutral prompts each share the same definition, source word, source donor, and evaluated state across one same-article and one cross-article target; screening used only tokenization, local efficacy, and mediator support, never route sign. At the fixed primary setting $\ell = 17$, $s = 1$, $\tau = 1$, mean R changes from -0.032 [-0.057, -0.010] within class to 0.087 [0.046, 0.132] across classes. The paired shift is 0.120 [0.077, 0.161]: 13/14 families shift in the predicted direction (one-sided binomial sign test $p = 0.00092$); an exact paired sign-flip randomization test using the paired magnitudes gives $p = 0.000122$. Eight of 14 individually show the strict sign reversal $R_{\text{within}} < 0 < R_{\text{cross}}$. The mean paired interaction remains positive at all 12 tested strength-temperature settings. Thus crossing the article mediator’s equivalence boundary systematically shifts the balance of target-aligned influence toward the public route while holding semantic family and source realization fixed. Component decomposition shows that this shift is chiefly increased public influence: $\Delta P = 0.102$ [0.062, 0.142], while $\Delta H = -0.018$ [-0.039, -0.0003]. The target-article policy itself moves by an additional 0.099 [0.055, 0.142] in the cross arm, directly linking the public increase to mediator recruitment.

Across the initial independent-family assay, public-minus-private target influence is 0.170 [0.049, 0.296] across classes, whereas private-minus-public influence is 0.094 [0.060, 0.126] within class. All six cross-class and all eight within-class families have the predicted sign (Figure 4

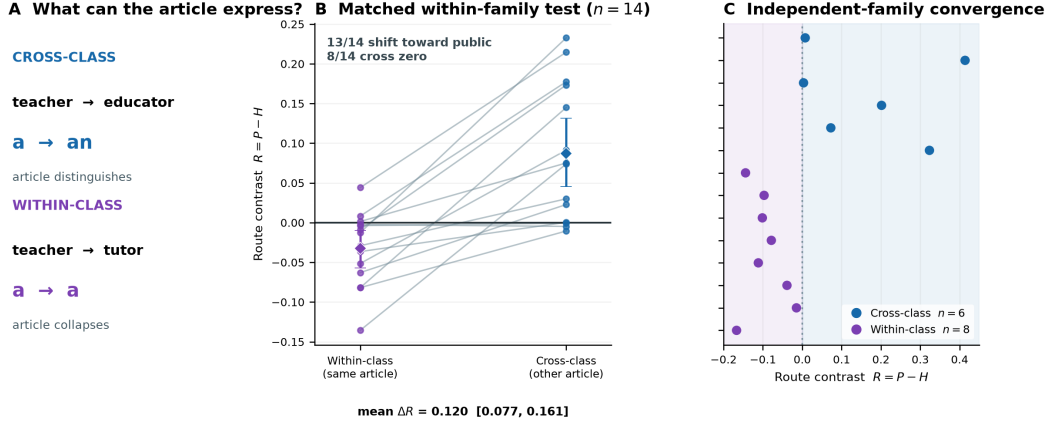


Figure 4: Mediator-relative route recruitment in Gemma 3 1B. A: the article can distinguish cross-class but not within-class alternatives. B: the matched assay compares within- and cross-article targets within each of 14 semantic families; lines share a neutral prompt, source state, and source word. Diamonds and bars show means and paired-bootstrap 95% intervals. C: the initial independent-family assay provides convergent evidence.

C). Their cross-minus-within interaction is 0.264 [0.137, 0.399], with exact family-label permutation $p = 0.00033$, providing convergent evidence across a broader set of lexical pairs.

The result supports a precise causal statement. Mediator distinguishability systematically shifts the balance of target-aligned causal influence toward the public route. When the article maps two alternatives to one equivalence class, token identity cannot carry their lexical difference; in the independent-family group means, the cross-class condition is public-dominant and the within-class condition is private-dominant. A generated token is therefore a lossy public projection relative to the downstream distinction of interest. We interpret the cross-class effect as selective recruitment of an additional public causal route, not conservation or transfer of a fixed information quantity between public and private channels. The experiment does **not** show that a natural online mechanism spontaneously chooses a carrier: the residual direction is explicitly constructed from named donors. The paired result identifies mediator-relative route recruitment under controlled interventions, not a sparse endogenous routing controller.

6 Closed-loop gain

Mediator visibility explains which distinctions can travel publicly, while Equation 1 explains their gain: local pressure changes a decoder policy, and continuation leverage determines the downstream consequence. Across 32 held-out 270M features, attribution moderately predicts local margin movement ($\rho = 0.530$), but attribution alone fails to predict public TV. Composing calibrated local pressure with intervention-off decoder susceptibility and branch leverage identifies and sizes the decisive-policy regime (aggregate $R^2 = 0.840$); prediction among small effects is weak and crossed feature-prompt uncertainty is wide $[-0.293, 0.986]$ (Figure A4). We therefore claim limited predictive validation of a mechanistically coherent 270M high-gain regime, not a universal predictor; full diagnostics are in Figure A5 and Figure A6.

7 Validity stress test at 1B

The sparse 1B campaign illustrates why the ladder’s rungs cannot be collapsed: the frozen top four are too weak, an effect-matched coalition leaves the binary article support, and a mediator-valid frontier yields only low-amplitude hybrid influence. Reporting only free generation would conflate these failure modes (full table in the appendix). We use the campaign as a validity demonstration, not evidence that scale reorganizes sparse planning mechanisms.

8 Discussion

The experiments support a two-part account of autoregressive causal organization.

Routing is mediator-relative. A public token does not expose the model’s entire future-relevant state. It projects that state into one vocabulary item. Relative to a target distinction, the projection may preserve information—as a/an does for consonant- versus vowel-initial alternatives—or collapse it, as a does for two consonant-initial synonyms. The neutral assay shows that this difference systematically shifts the balance of target-aligned causal influence toward the public route. This is stronger than observing that a token mediates a later effect: it connects route recruitment to what the mediator can distinguish.

Public routing has closed-loop gain. Once an intervention moves mediator policy, the written token is fed back through the model. Its effect is amplified when the decoder is near a policy boundary and the continuation branches have high leverage. S1 provides a high-gain, public-dominant case: a small change in intervention strength near the article-policy boundary changes the public action, which selects almost disjoint noun distributions. The large behavioral rewrite lives at the latent-to-symbol-to-continuation handoff, not in a similarly large fixed-token residual.

These principles reconcile results that otherwise appear contradictory. The S1 intervention can be selected by future-noun attribution yet transmit its downstream effect almost entirely through the article. Natural residual state can also influence a later noun under that same fixed article. And a constructed lexical direction can become public-dominant or private-dominant depending on whether article identity resolves its intended alternatives. Candidate content, local causal action, route, and behavioral magnitude are different questions.

This perspective sharpens mechanistic claims about planning, chain-of-thought, tool calls, delimiters, and scratchpads. Route describes transmission rather than cognitive function: a representation satisfying a latent-planning criterion can still affect a future token through a generated intermediate. An output token may be evidence of an upstream computation, a causal action, or both. If an intervention changes a later answer, replaying its intermediate output tests whether the output phenocopies the effect; restoring the baseline output tests whether it rescues behavior; fixed-output cells distinguish that public realization from influence that persists after the output is fixed. For larger mediator vocabularies, the same policy-weighted decomposition applies, with an omitted-mass bound for top- K approximations.

Our use of “public” and “private” is functional. Public state is readable from generated token identity and re-enters through the ordinary input channel. Private state is upstream contextual influence that remains after that identity is controlled. We do not localize a distinct hidden register, claim that the two routes are independent, or infer why the model computed the state. Indeed, public and private interventions interact, and fixed-token persistence may be distributed across attention paths.

The closest planning work already forces intermediate tokens and replays steered contexts [Hanna and Ameisen, 2026], distinguishes decodability from causal use [Ma and Rui, 2026], and demonstrates hidden influence beneath fixed scratchpad text [Shih et al., 2026]. The present synthesis adds a complete vector decomposition, mediator-support validity, a mediator-relative route reversal, and an empirically supported closed-loop gain account with limited predictive validation. Its methodological implication is simple: a future-token effect is not yet a future-state mechanism. One must show that the intervention works locally, that the mediator estimand remains valid, and how the closed loop transmits and amplifies the effect.

9 Scope, limitations, and conclusion

The sparse public-relay result concerns one Gemma 3 270M feature set, dictionary, and mediator. S1 is amplified $5\times$ and was not independently selected as a noun-specific semantic planning representation under Hanna and Ameisen’s criterion; we identify its route, not a unique natural ontology.

Full-residual assays establish causal capacity, not a natural sparse carrier or monotone scaling. Explicit synonym donors test routing, not spontaneous plan formation; only 8/14 families show strict sign reversal. Neither result establishes a routing controller or universality across models and mediators.

Greedy claims use exact conditional a/an mixtures, not unrestricted sampling; the binary estimand requires retained article support. The gain identity is algebraic: its 270M validation identifies the high-gain regime, but small-effect ranking, crossed feature–prompt precision, and 1B transfer are weak. Intervals describe finite prompt, feature, or family sets, not a task population.

References

- Emmanuel Ameisen, Jack Lindsey, Adam Pearce, et al. Circuit tracing: Revealing computational graphs in language models. *Transformer Circuits*, 2025. URL <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>.
- Lawrence Chan, Adrià Garriga-Alonso, Nicholas Goldowsky-Dill, Ryan Greenblatt, Jenny Nitishinskaya, Ansh Radhakrishnan, Buck Shlegeris, and Nate Thomas. Causal scrubbing: A method for rigorously testing interpretability hypotheses. *AI Alignment Forum*, 2022. URL <https://www.alignmentforum.org/posts/JvZhhzycHu2Yd57RN/causal-scrubbing-a-method-for-rigorously-testing>.
- Jacob Dunefsky, Philippe Chlenski, and Neel Nanda. Transcoders find interpretable LLM feature circuits. In *Advances in Neural Information Processing Systems*, 2024.
- Jaden Fiotto-Kaufman, Alexander R. Loftus, Eric Todd, et al. NNsight and NDIF: Democratizing access to open-weight foundation model internals, 2025. URL <https://arxiv.org/abs/2407.14561>.
- Atticus Geiger, Duligur Ibeling, Amir Zur, Maheep Chaudhary, Sonakshi Chauhan, Jing Huang, Aryaman Arora, Zhengxuan Wu, Noah Goodman, Christopher Potts, and Thomas Icard. Causal abstraction: A theoretical foundation for mechanistic interpretability. *Journal of Machine Learning Research*, 26:1–63, 2025.
- Gemma Team. Gemma 3 technical report, 2025. URL <https://arxiv.org/abs/2503.19786>.
- Google DeepMind. Gemma scope 2. Hugging Face, 2025. URL <https://huggingface.co/google/gemma-scope-2>.
- Michael Hanna and Emmanuel Ameisen. Latent planning emerges with scale, 2026. URL <https://arxiv.org/abs/2604.12493>.
- Michael Hanna et al. circuit-tracer: A library for finding feature circuits. GitHub, 2025. URL <https://github.com/decoderresearch/circuit-tracer>.
- Nicole Ma and Nick Rui. Where’s the plan? locating latent planning in language models with lightweight mechanistic interventions, 2026. URL <https://arxiv.org/abs/2605.07984>.
- Aaron Mueller, Jannik Brinkmann, Millicent Li, Samuel Marks, Koyena Pal, Nikhil Prakash, Can Rager, Aruna Sankaranarayanan, Arnab Sen Sharma, Jiuding Sun, Eric Todd, David Bau, and Yonatan Belinkov. The quest for the right mediator: Surveying mechanistic interpretability for NLP through the lens of causal mediation analysis. *Computational Linguistics*, 52(1):331–378, 2026. doi: 10.1162/coli.a.572. URL <https://aclanthology.org/2026.cl-1.10/>.
- Debjit Paul, Robert West, Antoine Bosselut, and Boi Faltings. Making reasoning matter: Measuring and improving faithfulness of chain-of-thought reasoning. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pages 15012–15032, 2024.
- Benjamin Shih, John Winnicki, and Eric Darve. Do models read what they write? causal registers in scratchpad reasoning, 2026. URL <https://arxiv.org/abs/2606.29522>.
- Jesse Vig, Sebastian Gehrmann, Yonatan Belinkov, Sharon Qian, Daniel Nevo, Yaron Singer, and Stuart Shieber. Investigating gender bias in language models using causal mediation analysis. In *Advances in Neural Information Processing Systems*, 2020.

A Supplementary methods and results

A.1 Reproducibility map

The repository treats manuscript prose as a view over frozen machine-readable artifacts. The principal claims can be recomputed from the following experiment directories.

Claim	Experiment artifact
Six-cell S1 split	<code>experiments/s1_article_factorial/results/</code>
S1 planning-feature audit	<code>experiments/s1_planning_feature_audit/results/audit.json</code>

Claim	Experiment artifact
Replay and rescue	experiments/s1_replay_rescue/results/
Decoder boundary and precision	experiments/boundary_precision_policy/results/
Private residual references	experiments/natural_residual_carrier_regimes/results/ and experiments/cross_scale_normalized_residual/results/
Neutral synonym routing	experiments/neutral_synonym_repair/results/
Matched semantic-triad routing	experiments/matched_semantic_triads_repaired/results/
Matched component and policy audits	experiments/matched_semantic_triads_repaired/results/component_diagnostic.json and experiments/matched_semantic_triads_repaired/results/article_policy_audit.json
Independent Figure 4 recomputation	experiments/neutral_synonym_repair/results/independent_recomputation.json
Closed-loop gain	experiments/closed_loop_gain_law/results/
Sparse 1B stress tests	experiments/gemma_1b_effect_matched/results/ and experiments/gemma_1b_sparse_frontier/results/

407 All summary JSON files record model identifiers, sample sizes, interval methods, and primary set-
408 tings. Plotting scripts read those artifacts rather than values copied from manuscript prose. The
409 270M model snapshot is 9b0cfec892e2bc2afd938c98eabe4e4a7b1e0ca1; the affine Gemma
410 Scope 2 dictionary snapshot is b218cd5d69dc2fa71cff448b68d625e6c9702d49 with configura-
411 tion width_262k_10_medium_affine. Discussed decoder/encoder vectors and selection metadata
412 are archived under experiments/archived_feature_vectors/. Experiments ran on one Apple
413 M1 workstation with 16 GB RAM and no external GPU or cluster compute. Individual timed runs
414 are recorded in their result summaries or logs. Summing those records, including the six-hour 1B
415 attribution campaign, gives approximately 13 workstation-hours; this is a lower bound because some
416 early exploratory and failed runs predated systematic timing.

417 The frozen S1 audit checks its relationship to the planning-feature motivation without rerunning
418 inference. Each of the four selected features is active at the pre-article position and has positive
419 direct attribution to the intended future noun on all eight selection prompts. Together with the
420 intervention results, this establishes future-noun-attributed pre-article causal handles. It does not
421 establish noun-specific semantic identity: the same features recur across distinct future nouns, and S1
422 selection jointly required positive article attribution. We therefore retain the term *attribution-selected*
423 *S1 features* rather than claiming an independently selected Hanna-style semantic planning feature.

424 A.2 Exact six-cell measurements

425 For each fixed article, the intervention remains active at its original pre-article position during noun
426 prediction. This differs from deleting the intervention after the article is inserted. We compute
427 full-sequence forwards with `use_cache=False`; multi-token noun likelihoods are teacher-forced
428 one token at a time. Full-vocabulary TV is $TV(p, q) = \frac{1}{2} \sum_v |p_v - q_v|$. We additionally record
429 Jensen–Shannon divergence, top- k overlap, target-minus-source logits for prespecified lexical twins,
430 vector cosine, and $\|E_{\text{total}} - E_{\text{public}} - E_{\text{private}}\|_1$.

431 For stochastic mediator support $\mathcal{B}$, the complete policy-weighted decomposition is

$$\begin{aligned}
E_{\text{public}} &= \sum_{b \in \mathcal{B}} [q_1(b) - q_0(b)] Y(0, b), \\
E_{\text{private}} &= \sum_{b \in \mathcal{B}} q_1(b) [Y(1, b) - Y(0, b)], \\
E_{\text{total}} &= E_{\text{public}} + E_{\text{private}}.
\end{aligned}$$

432 The S1 decomposition over all 20 held-out prompts is:

Quantity	Mean	95% interval
Total noun TV	0.893	[0.784, 0.969]
Treated-token replay TV	0.891	[0.776, 0.969]
Matched treated-token leftover TV	0.028	[0.021, 0.035]
Replay-total cosine	0.950	[0.903, 0.979]

433 Nineteen prompts switch article and noun. On that descriptive switched subset, total, replay, and
434 leftover TVs are 0.940, 0.937, and 0.029. We use the all-prompt values in the headline to avoid
435 post-selection.

436 A.3 Boundary and stochastic-policy audits

437 Native model, CLT, and residual tensors are bfloat16. A diagnostic head casts the final normalized
438 residual and unembedding to float32 without rerunning the transformer in float32. Across three
439 prespecified prompts, a search of width 2.44×10^{-4} still produces 0.992 free-path TV while fixed a
440 and an move 0.009 and 0.004. The audit shows that native 0/0.125-logit steps reflect final-readout
441 quantization, while the continuation discontinuity under greedy article choice is real.

442 For stochastic policies we renormalize over a/an and mix the two fixed-article noun distributions.
443 Retained article mass exceeds 0.998 at low temperature in the boundary assay. Policy-vector alignment
444 with the total remains at least 0.99 across tested temperatures; the private component remains small.
445 These calculations evaluate the exact conditional mixture and introduce no sampling variance. The
446 strength-boundary and temperature sweeps make the same point visually: the total effect tracks the
447 public component while the fixed-token leftover stays small (Figure A1).

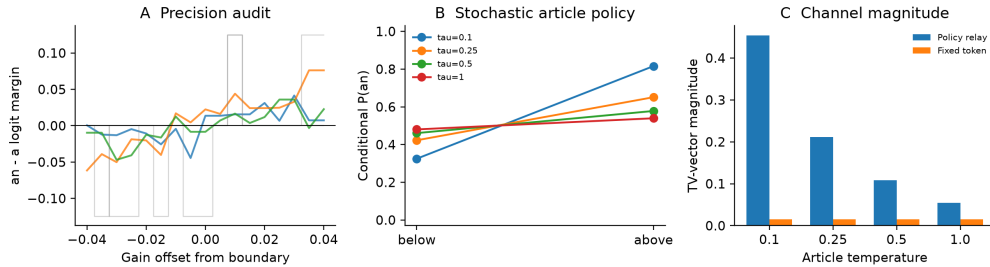


Figure A1: S1 boundary and stochastic-policy robustness. A: a float32 final-head audit resolves the article margin more finely around the boundary. B: softening the article policy turns the discrete switch into graded probability movement. C: total influence shrinks with temperature, with the public component tracking it and the fixed-token residual remaining small.

448 A.4 Private-route controls and scale

449 Natural-state-derived patches are constructed from target-uncued descriptions. The pooled develop-
450 ment sweep selects decoder layer 12 at 270M and layer 14 at 1B. At each scale and route condition
451 we report raw target-minus-source movement, fraction of the baseline lexical gap closed, target-odds
452 multiplier, patch norm relative to natural residual RMS, and stochastic public/private TVs. Controls
453 include a semantic wrong target and sign reversal. The target effect is positive at both scales, while
454 normalized comparisons do not support a monotone increase at 1B. The fuller normalized comparison
455 in Figure A2 shows both the cross-scale replication and the absence of a monotone scale trend; the
456 control directions remain much weaker than the intended target direction.

457 The stronger dense residual reference patches the decoder-layer output at the inserted article position.
458 Four development pairs select layer 12; four held-out same-article pairs yield 6.44 target-specific
459 logits versus 0.14 for ten matched-norm random directions. This is a reachability and stack-sensitivity
460 control, not evidence that S1 represents the same lexical direction.

461 A.5 Neutral synonym assay details

462 The bare neutral template ending immediately after the definition failed mediator-support screening
463 before route inference: called frequently outranked both articles, mean a/an mass was approxi-

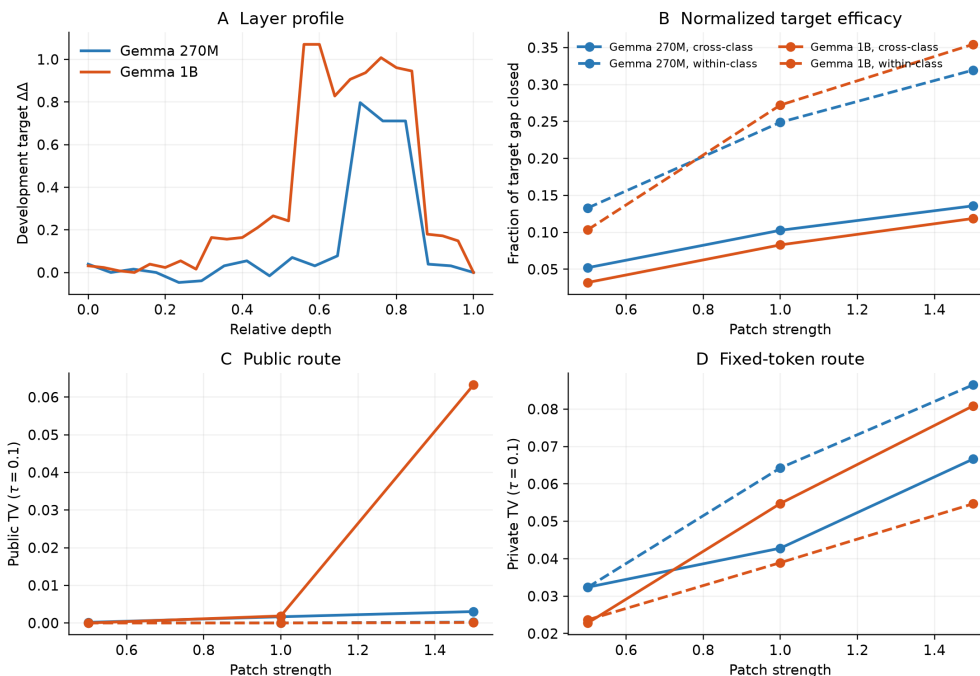


Figure A2: Full cross-scale private-route comparison. A: development layer profiles. B: normalized target efficacy across patch strengths. C–D: public and fixed-token effect magnitudes across strength. The target effect is present at both Gemma 3 270M and 1B, but normalization does not support a monotone increase with scale. Specificity controls appear in Figure 3.

mately 0.27, and only 7/14 prompts had an article top token. The frozen evaluated format therefore uses one unrelated demonstration:

Someone who studies living organisms is a biologist. [definition] is

Donor prompts append Use exactly the word [word] to the same format. The demonstration names neither tested synonym, initial letter, nor tested article class. Leave-one-family-out development selection chooses layer 17 in all folds. All 14 directions have positive local efficacy on the neutral evaluated prompt, despite direct donor separation being positive in 13/14.

At $s = 1, \tau = 1$, the main route quantities are:

Regime	Public TV	Private TV	Public target effect	Private target effect
Cross-class (N=6)	0.176	0.021	0.189	0.019
Within-class (N=8)	0.009	0.061	-0.001	0.093

Both signed simple effects have the predicted direction under exact sign flips, and the independent family-label interaction has exact two-sided $p = 0.00033$ over 3,003 assignments. The interaction remains positive for all 12 strength-temperature settings (Figure A3). This visual check shows that the sign reversal is not peculiar to one chosen strength or article-policy temperature. We archive the rejected bare-neutral output under `results_bare_neutral_invalid_support/` rather than silently replacing it.

The repaired paired extension freezes 22 candidate semantic triads before screening. Fourteen pass the route-blind tokenization, local-efficacy, and article-support criteria. Each retained definition is evaluated from the same source word and neutral prompt with one same-article and one cross-article

target. At $s = 1, \tau = 1$, within-class $R = P - H$ is $-0.032 [-0.057, -0.010]$, cross-class R is $0.087 [0.046, 0.132]$, and their paired difference is $0.120 [0.077, 0.161]$. Thirteen of 14 paired shifts are positive (one-sided binomial sign test $p = 0.00092$; exact paired sign-flip randomization $p = 0.000122$); eight have the strict double-dissociation sign. The paired interaction remains positive at every combination of three strengths and four temperatures. This is the primary control for semantic-family composition in the mediator-relative claim.

The component decomposition makes the source of this route shift explicit. Within class, mean public and private target effects are respectively $-0.0017 [-0.0108, 0.0097]$ and $0.0307 [0.0142, 0.0509]$; across classes they are $0.0998 [0.0581, 0.1446]$ and $0.0127 [0.0041, 0.0217]$. Consequently, the paired changes are $\Delta P = 0.1015 [0.0625, 0.1423]$ and $\Delta H = -0.0180 [-0.0389, -0.0003]$: the larger public component accounts for most of the change in R , with a smaller reduction in private influence. A deterministic 28-cell audit recovers the missing conditional article probabilities and exactly reconstructs every frozen public effect (maximum absolute error zero). Mean target-article policy movement is 0.0007 within class and 0.0993 across classes; their paired difference is $0.0986 [0.0549, 0.1418]$, positive in 12/14 families. Thus the public increase accompanies stronger recruitment of the target article, not continuation leverage alone.

The outcome-blind potency audit does not find stronger cross-article interventions. Mean cross versus within direction norms are 699 versus 761 (paired difference $-62 [-188, 60]$); fixed-article local efficacies are 0.353 versus 0.415 (difference $-0.063 [-0.174, 0.063]$); and treated article support is 0.912 versus 0.913 . Cross targets do have a higher baseline target-minus-source lexical gap (paired difference 7.01 logits), so target-word identity is not held fixed; this is disclosed rather than interpreted as route evidence. Familywise gap differences do not predict the paired route shift (Pearson $r = -0.005$, Spearman $\rho = -0.015$, 100,000-permutation two-sided $p = 0.986$); leave-one-family-out Pearson correlations range from -0.104 to 0.148 . Because neither direction norm nor fixed-mediator causal potency favors the cross arm, no outcome-blind efficacy rescaling is triggered.

All 22 candidates and their route-blind screening outcomes are shown below. Semantic validity was assessed during pre-model curation before this candidate set was frozen and before tokenization, model screening, or route computation.

Family	Source	Same-article target	Cross-article target	Retained	Route-blind failure reason
teaching	a teacher	a tutor	an educator	yes	—
legal	a lawyer	a counsel	an attorney	no	cross local efficacy tokenization
medicine	a doctor	a physician	an internist	no	—
writing	a writer	a novelist	an author	yes	—
performance	an actor	an entertainer	a performer	yes	—
assistance	an aide	an assistant	a helper	no	within/cross treated support
events	a coordinator	a planner	an organizer	yes	—
leadership	a leader	a boss	an executive	yes	—
academia	a researcher	a scientist	an academic	yes	—
agriculture	a farmer	a grower	an agriculturist	no	tokenization
business	a founder	a businessperson	an entrepreneur	no	tokenization
art	an artist	an illustrator	a painter	yes	—
police	a cop	a policeman	an officer	yes	—
speaking	a speaker	a presenter	an orator	yes	—
employment	a worker	a laborer	an employee	yes	—
ownership	a proprietor	a holder	an owner	yes	—
invention	an inventor	an innovator	a creator	yes	—

Family	Source	Same-article target	Cross-article target	Retained	Route-blind failure reason
students	a student	a pupil	an under-graduate	yes	–
driving	a driver	a motorist	an operator	no	cross local efficacy
aviation	an aviator	an airman	a pilot	no	tokenization
accounts	an accountant	an auditor	a bookkeeper	no	tokenization
management	a manager	a director	an executive	yes	–

509 An independent, artifact-only recomputation imports none of the experiment’s analysis or plotting
510 code. Starting from frozen row-level outputs, it reproduces all 14 local-efficacy checks, article-support
511 statistics, per-family route effects, 6/6 and 8/8 original signs, bootstrap summaries, exact sign-flip
512 tests, the 0.264 independent-group interaction, and its exact $p = 0.00033$ label permutation test.
513 The same independent runner also reconstructs all 22 triad eligibility decisions, the exact retained
514 set, every paired route contrast, both headline tests, and all 12 robustness settings. It fails if any
515 recomputed headline differs from the frozen summary beyond 10^{-12} .

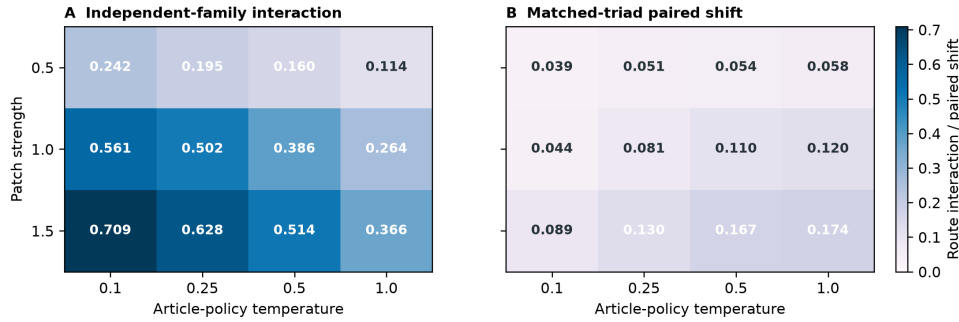


Figure A3: Synonym-route robustness. A: the initial independent-family interaction. B: the matched-triad paired shift. Every strength–temperature cell is positive in both assays.

516 A.6 Closed-loop gain validation

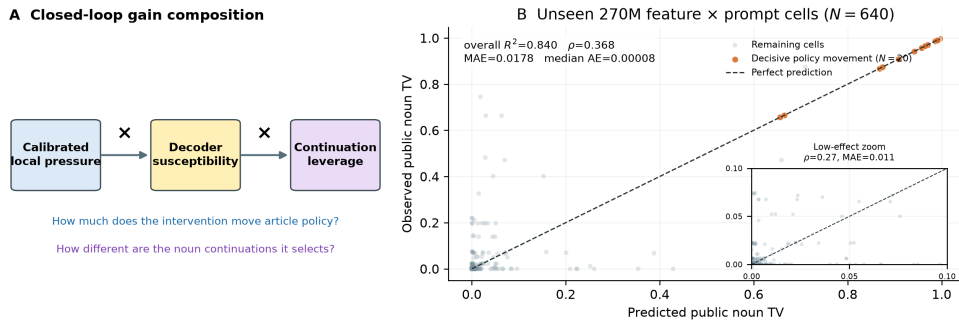


Figure A4: Closed-loop gain and held-out prediction at Gemma 3 270M. A: attribution is calibrated to local causal pressure, then composed with decoder susceptibility and continuation leverage. B: the composition captures the high-gain regime; orange cells have both observed and predicted article-policy movement above 0.9, while the inset resolves the low-effect cloud. Holdout excludes feature and prompt identity from causal-pressure calibration; intervention-off decoder susceptibility and branch leverage of the held-out prompt are permitted observed inputs. Aggregate R2 is tail-sensitive and is not a claim of uniformly precise cell-level prediction.

517 The two-branch identity holds to numerical precision. Prediction fits a mapping from attribution
518 to local article-margin movement on training features and prompts, then evaluates public TV after
519 composition with intervention-off susceptibility and branch leverage. Combined feature-and-prompt
520 holdout results for 270M are:

Temperature	Attribution only R^2	+ susceptibility	Full composition
0.10	-0.547	0.827	0.840
0.25	-0.529	0.861	0.875
0.50	-0.191	0.857	0.870
1.00	0.395	0.851	0.864

521 At $\tau = 0.1$, constant, susceptibility-only, leverage-only, and activation baselines have $R^2 =$
522 $-0.052, -0.082, -0.053, -0.045$. The oracle using the measured margin and full leverage has
523 $R^2 = 1$ by identity and is not a prediction. At 1B, combined-holdout transfer is weak; we report
524 that failure rather than pool scales. Figure A5 and Figure A6 show the complete diagnostic set:
525 attribution has moderate local calibration (A), fixed-token attribution correlations are inconclusive
526 (B), the full 270M composition outperforms simple baselines (C), bin means expose calibration
527 across the dynamic range (D), tail removal exposes the aggregate R^2 's dependence on large effects
528 (E), and temperature and scale checks delimit transfer (F).

529 The calibration intervals in panel D use 10,000 crossed bootstrap resamples: feature and prompt
530 identities are sampled independently with replacement and their multiplicities are crossed into cell
531 weights. The same blocked procedure gives aggregate $R^2 = 0.840 [-0.293, 0.986]$ and MAE
532 $0.0178 [0.0036, 0.0377]$. Panel E reports MAE alongside tail-trimmed R^2 : removing large effects
533 shrinks absolute error but collapses outcome variance even faster, making R^2 strongly negative
534 without implying large absolute mistakes.

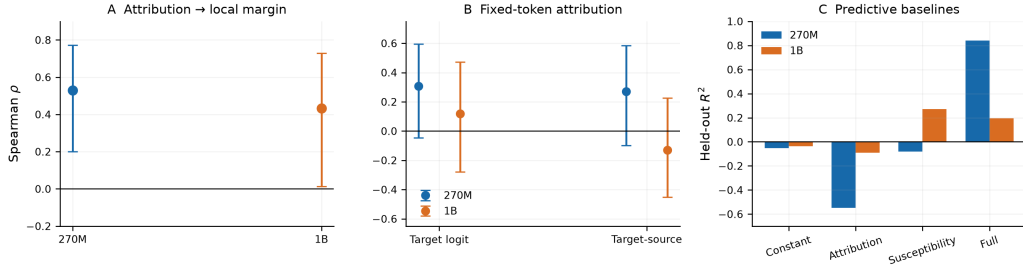


Figure A5: Closed-loop gain diagnostics I. A: attribution-to-local-efficacy calibration. B: fixed-token target correlations with broad intervals. C: predictive baselines.

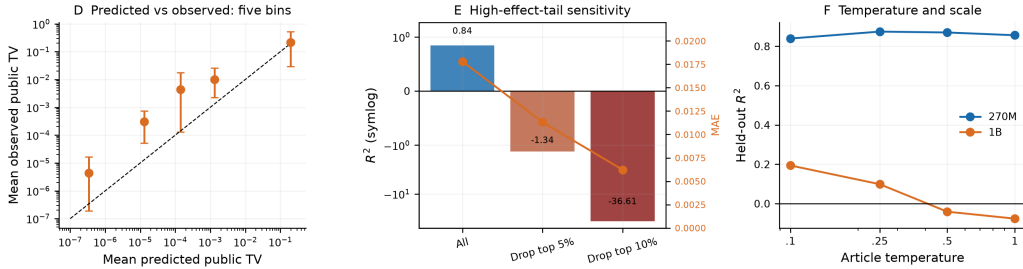


Figure A6: Closed-loop gain diagnostics II. D: equal-count predicted-versus-observed calibration with crossed feature-prompt bootstrap intervals. E: R^2 and MAE respond differently when the largest observed effects are excluded because the remaining outcome variance is tiny. F: temperature robustness at 270M and weak transfer for available 1B handles.

535 A.7 Sparse 1B ladder failures

Sparse handle	Local efficacy	a/an support	What follows?
Frozen top four	No: 0.075-logit mean movement	Yes	No route claim
Effect-matched 24	Yes	No: called often dominates	No binary route claim
Constrained frontier	Yes	Yes: 20/20 prompts	Low-amplitude hybrid route

The original four-feature sparse intervention was independently selected from 1,200-node attribution graphs but moved the held-out article margin only 0.075 logits. A 24-feature effect-matched coalition increased local movement but made non-article tokens such as called dominate, invalidating the binary mediator estimand. A constrained frontier feature at layer 18, feature 5015, preserved article support on 20/20 prompts. At $7.5\times$ it changed only one greedy article, but its exact stochastic decomposition showed low-amplitude hybrid influence. Reconstruction audits reveal uneven CLT fidelity and a substantial affine-skip contribution; graph-cap audits make simple truncation a less likely sole explanation. These results motivate the ladder but do not establish a scale-dependent reorganization.

A.8 Negative and superseded studies

S2 frequently leaves the intended article domain and is retained only as a forensic warning. A sparse layer-5 feature produces exploratory fixed-token movement but has weak matched controls and may encode vowel class. The released is/are subtraction task fails behavioral and local-efficacy prerequisites in both models, so it cannot test mediator capacity. Earlier first-letter synonym prompts are superseded because the evaluated source cue contradicted the target synonym. An arbitrary A/B few-shot pilot does show that context can alter target nouns with the code fixed (effects 0.224–0.245 across reliability banks), but the visible context itself remains available and the task is less mechanistically clean than the neutral synonym assay; it is not part of the main argument.

A.9 Responsible use and checklist notes

The study uses public pretrained models and synthetic prompts. No personal data, human subjects, or newly collected sensitive data are involved. Compute was run locally on Apple silicon. Code, frozen configurations, checkpoints, invalidated outputs, and machine-readable summaries are retained for release after review; the anonymous submission does not include the working repository because its frozen logs and metadata contain identifying local paths. Confidence intervals, exact tests, sample sizes, development/evaluation separation, and known post-selection are stated with each claim.

The principal positive impact is improving the validity of mechanistic claims about model computation. A potential negative impact is overconfidence in mechanistic interpretations if route-identification assumptions are ignored or generalized beyond their validated scope; this work has no direct deployment path.

A.10 Assets and licenses

Asset	Version or snapshot	License or terms
Gemma 3 270M/1B PT	9b0cf...e0ca1; fcf18...8eb29	Gemma Terms of Use
Gemma Scope 2 PT CLTs	b218c...02d49; b738d...e873	CC BY 4.0
circuit-tracer	0.5.0	MIT
Transformers	4.57.3	Apache-2.0
NNsight	0.7.0	MIT
Hanna–Ameisen prompts	commit 993cff5...8dab	MIT

NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

Answer: [Yes]

Justification: The abstract and Section 1 distinguish the scoped S1 relay-and-amplification result from private-route capacity. Support includes the six-cell split, replay/rescue, precision and stochastic-policy audits, natural residual controls, a paired 14-family mediator-relative assay, held-out attribution calibration, and disjoint development/test patches.

Guidelines:

- The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
- The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
- The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
- It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [Yes]

Justification: Section 9 scopes the evidence to one sparse handle, dictionary, and grammatical mediator; distinguishes full-residual capacity interventions from natural sparse mechanisms; rejects a monotone scaling claim; identifies donor-derived rather than spontaneous routing; reports that only 8/14 matched families show strict sign reversal; notes the absence of cross-model and cross-mediator replication; and limits the binary conditional-policy and gain-prediction claims.

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [N/A]

Justification: The paper uses an exact probability-vector identity and a causal identification observation, but claims no theorem requiring a formal proof.

Guidelines:

- The answer [N/A] means that the paper does not include theoretical results.
- All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
- All assumptions should be clearly stated or referenced in the statement of any theorems.
- The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
- Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
- Theorems and Lemmas that the proof relies upon should be properly referenced.

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Sections 3–5 and the appendix specify model, transcoder, package versions, protocols, metrics, and experiment directories with configs and summaries. An independent artifact-only runner recomputes both Figure 4 assays, including all 22 paired-candidate eligibility decisions, support checks, intervals, and exact tests, without importing the original analysis code.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general, releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
 - (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
 - (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
 - (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).

- (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. Open access to data and code

Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [No]

Justification: The paper and supplement provide exact model and package snapshots, frozen splits, sample sizes, intervention definitions, statistical procedures, and a claim-to-artifact reproducibility map. The working repository is withheld during anonymous review because frozen logs and metadata contain identifying local paths; a sanitized code and result release will follow review.

Guidelines:

- The answer [N/A] means that paper does not include experiments requiring code.
- Please see the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
- The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (<https://neurips.cc/public/guides/CodeSubmissionPolicy>) for more details.
- The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
- The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
- At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. Experimental setting/details

Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

Answer: [Yes]

Justification: Section 3 and the appendix give model versions, prompts, frozen splits, layer/strength selection, intervention procedures, and evaluation metrics.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. Experiment statistical significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [Yes]

Justification: Main factorial and reference effects use 10,000-resample prompt- or pair-level nonparametric bootstrap intervals; the matched-triad result additionally reports an exact one-sided binomial sign test and an exact paired sign-flip randomization test. Random seeds are averaged within pair before pair-level inference.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
- The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
- The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
- The assumptions made should be given (e.g., Normally distributed errors).
- It should be clear whether the error bar is the standard deviation or the standard error of the mean.
- It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
- For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
- If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. Experiments compute resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Experiments ran on one Apple M1 workstation (16 GB) using CPU inference with streamed transcoder loads and no external GPU or cluster compute. Individual timed runs are recorded in summaries or logs; their sum is approximately 13 workstation-hours, a lower bound because some early exploratory and failed runs predated systematic timing (Appendix A.1).

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. Code of ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://neurips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: The work analyzes open-weight models on synthetic grammatical prompts and conforms to the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.

- 779 • If the authors answer [No], they should explain the special circumstances that require a
780 deviation from the Code of Ethics.
781 • The authors should make sure to preserve anonymity (e.g., if there is a special consid-
782 eration due to laws or regulations in their jurisdiction).

783 10. Broader impacts

784 Question: Does the paper discuss both potential positive societal impacts and negative
785 societal impacts of the work performed?

786 Answer: [Yes]

787 Justification: Appendix A.9 describes the positive methodological impact, the risk of
788 overconfident mechanistic interpretation, and the absence of a direct deployment path.

789 Guidelines:

- 790 • The answer [N/A] means that there is no societal impact of the work performed.
791 • If the authors answer [N/A] or [No], they should explain why their work has no societal
792 impact or why the paper does not address societal impact.
793 • Examples of negative societal impacts include potential malicious or unintended uses
794 (e.g., disinformation, generating fake profiles, surveillance), fairness considerations
795 (e.g., deployment of technologies that could make decisions that unfairly impact specific
796 groups), privacy considerations, and security considerations.
797 • The conference expects that many papers will be foundational research and not tied
798 to particular applications, let alone deployments. However, if there is a direct path to
799 any negative applications, the authors should point it out. For example, it is legitimate
800 to point out that an improvement in the quality of generative models could be used to
801 generate Deepfakes for disinformation. On the other hand, it is not needed to point out
802 that a generic algorithm for optimizing neural networks could enable people to train
803 models that generate Deepfakes faster.
804 • The authors should consider possible harms that could arise when the technology is
805 being used as intended and functioning correctly, harms that could arise when the
806 technology is being used as intended but gives incorrect results, and harms following
807 from (intentional or unintentional) misuse of the technology.
808 • If there are negative societal impacts, the authors could also discuss possible mitigation
809 strategies (e.g., gated release of models, providing defenses in addition to attacks,
810 mechanisms for monitoring misuse, mechanisms to monitor how a system learns from
811 feedback over time, improving the efficiency and accessibility of ML).

812 11. Safeguards

813 Question: Does the paper describe safeguards that have been put in place for responsible
814 release of data or models that have a high risk for misuse (e.g., pre-trained language models,
815 image generators, or scraped datasets)?

816 Answer: [N/A]

817 Justification: The study introduces no new model or scraped dataset and therefore creates no
818 release-specific high-risk asset.

819 Guidelines:

- 820 • The answer [N/A] means that the paper poses no such risks.
821 • Released models that have a high risk for misuse or dual-use should be released with
822 necessary safeguards to allow for controlled use of the model, for example by requiring
823 that users adhere to usage guidelines or restrictions to access the model or implementing
824 safety filters.
825 • Datasets that have been scraped from the Internet could pose safety risks. The authors
826 should describe how they avoided releasing unsafe images.
827 • We recognize that providing effective safeguards is challenging, and many papers do
828 not require this, but we encourage authors to take this into account and make a best
829 faith effort.

830 12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: We cite Gemma 3, Gemma Scope 2, circuit-tracer, NNsight, and the source prompt repository. Appendix A.10 names licenses and terms (Gemma Terms of Use; Gemma Scope 2 CC BY 4.0; MIT; Transformers Apache-2.0), pinned snapshots, package versions, and the MIT-licensed Hanna–Ameisen repository commit and checksum.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, paperswithcode.com/datasets has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset’s creators.

13. New assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: Experiment configs, scripts, and JSON summaries are documented in the repository READMEs and appendix map.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. Crowdsourcing and research with human subjects

Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

Answer: [N/A]

Justification: No crowdsourcing or human-subjects research.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.

- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

Answer: [N/A]

Justification: No human-subjects research.

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
- We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
- For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

Answer: [Yes]

Justification: AI agents assisted with literature synthesis, experimental-design brainstorming and critique, implementation, execution, result aggregation, and manuscript editing. The authors made the final scientific decisions and independently verified the reported experiments, statistics, figures, and citations against frozen artifacts and source papers.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.